

CorneaNet: A Deep Learning Framework for Explainable and Early Detection of Keratoconus from Multimodal Corneal Tomography

Deepa. R

Department of Artificial Intelligence
and Machine Learning
St. Joseph's College of Engineering
Chennai, India
deepa@st.josephs.ac.in

Dr. R. Niruban

Department of Electronics and
Communication Engineering
St. Joseph's College of Engineering
Chennai, India
deepa@st.josephs.ac.in

S. Ebenazer Roselin

Assistant professor , Department of
CSE(Cyber Security)
Sri Sairam Institute of Technology
Chennai, India
ebenazer.sc@sairamit.edu.in

Dr. N.Vinitha

Department of Computer and
Communication Engineering
Sri Sairam Institutions
Chennai, India
vinitha.cce@sairamit.edu.in

Abstract—Keratoconus (KC) is a progressive ocular disease involving thinning and conical steepening of the cornea, which may result in serious visual impairment if left undiagnosed or untreated. It is crucial to diagnose keratoconus early and correctly in order to take appropriate action and prevent the condition from worsening. Traditional diagnostic methods, are slow to respond to new forms of disease, and much time and effort are spent in this process. Although current deep learning (DL) methods have shown high performance in computerized KC detection. This paper introduces CorneaNet, a new multi-input convolutional neural network (CNN) structure for explainable and robust keratoconus classification from Scheimpflug tomography scans. Our model takes four multimodal inputs: anterior curvature, posterior elevation, pachymetry, and wavefront aberration maps. Gradient-weighted Class Activation Mapping (Grad-CAM) is integrated into the framework for creating visual heatmaps that identify the pathological features. Trained and tested on a large 12,570 scans from 1,845 patients dataset, CorneaNet attained area under the curve (AUC) of 0.997 for the detection of clinical KC and 0.983 for detecting subclinical KC on a held-out test set. The Grad-CAM outputs corroborated that the attention of the model is consistent with established clinical biomarkers.

Keywords— Scheimpflug tomography, CorneaNet, keratoconus, Gradient-weighted Class Activation Mapping, biomarkers

I. INTRODUCTION

Keratoconus (KC) is a progressive eye disease that changes the shape and structural integrity of the cornea, the transparent outer covering of the eye. The cornea is responsible for bringing light into focus on the retina, as it covers and guards important internal eye structures. The outer epithelium is a protective covering, Bowman's layer supports, the stroma creates transparency through ordered collagen, Descemet's membrane keeps the shape, and the endothelium maintains fluid balance [1]. In combination, these layers allow the cornea to bend light and guard the inner structures of the eye. The corneal structure is illustrated in figure1.

Keratoconus is a degenerative eye disease in which normally dome-shaped cornea thins and bulges outward into a conical shape. Daily functions such as driving and reading are difficult because of the blurred and distorted vision generated by this abnormality. The disease usually starts in

adolescent years and can continue to its mid-30s. It occurs in about 1 in 2,000 people around the world [2].

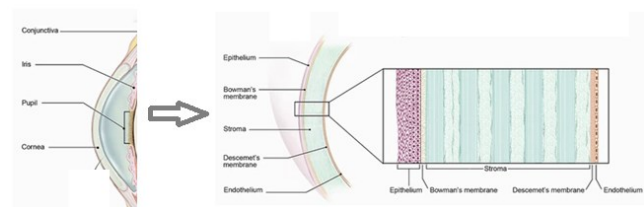


Figure 1: corneal structure

Symptoms of keratoconus are unsightly or distorted vision, increased light sensitivity, and frequent changes in prescription glasses. Soft contact lenses or glasses may be employed to manage the initial stages. Specialist contact lenses or surgical options such as corneal transplantation may be necessary when the condition deteriorates. [3]. Ongoing monitoring by an ophthalmologist is vital for proper control and maintenance of vision. Asymmetrical astigmatism, usually difficult to correct, most commonly follows corneal thinning and may produce progressive visual loss. [4]. For the detection of keratoconus, corneal topography and corneal tomography are the most commonly employed methods of investigation [5].

Early recognition and severity staging of KC is critical in determining treatment, estimating the course of the disease, maximizing results, and providing the best possible care at every stage of the disease. By giving a clear image of how severe the disease is, staging permits more accurate management, enhancing both short-term visual results and long-term quality of life for patients with KC. This has previously been accomplished through various methodologies utilized in identifying KC into multiple stages using ML algorithms [6]. For example, research studies [7] and staged KC into five grades (I-V) according to corrected distance visual acuity (CDVA) [8]. Equally, the authors of [9] used Amsler-Krumeich (AK) classification system to classify corneas into four levels (1–4). The AK system mainly utilizes keratometry but also incorporates other parameters like refraction and pachymetry to provide a more thorough evaluation of corneal deformation [10].

On the contrary, the writers of [11] suggested a classification model, categorizing KC corneas into mild and moderate only using a framework they created. The inconsistency between studies indicates the unavailability of a standard that is widely accepted for grading the severity of KC. This difference in classification techniques can be attributed to differences in diagnostic criteria utilized, different imaging modalities, and the absence of publicly available data sets that can be used as a benchmark for KC severity prediction.

Deep learning [12] has transformed medical imaging by increasing precision and efficiency in image analysis, and patient outcomes have improved as a result. Deep learning methods can identify and diagnose diseases with incredible accuracy through the use of enormous datasets. Medical photographs are improved in quality through the application of deep learning algorithms for easier visualization and interpretation, which is important for proper diagnosis.

Nonetheless, a key impediment to clinical adoption of these influential models is that they are "black-box." Typical DL models are not capable of providing a justification or visual explanation for a prediction, and in a clinic where having a rationale for a diagnosis is of utmost importance for assurance and safety of the patient [13], this is unacceptable. In addition, most current methods work with a single type of image alone, not exploiting the full multimodal information generated by new tomographers, which might be more robust and accurate.

To overcome these limitations, CorneaNet, a new explainable deep learning architecture is introduced for the early and robust detection of keratoconus. The main contributions of this paper are fourfold:

- A new multi-input CNN architecture that jointly processes four different corneal tomographic maps like anterior curvature, posterior elevation, pachymetry, and wavefront aberration to facilitate an end-to-end data-driven evaluation.
- The lossless integration of an explainability mechanism with Gradient-weighted Class Activation Mapping (Grad-CAM), yielding visual heatmaps that identify the most important areas to the model's prediction.
- A thorough testing on a large, clinically validated dataset showcasing state-of-the-art performance not only in the detection of clinical KC but, more significantly, in the identification of difficult subclinical cases.
- A demonstration that the visual explanations produced by the model match recognized clinical biomarkers of KC, thus closing the loop between AI inference and clinical knowledge.

The rest of this paper is organized as follows: Section II summarizes related work. Section III describes the dataset, preprocessing, and the proposed CorneaNet architecture. Section IV shows the experimental results and analysis. Section V presents the clinical implications, limitations, and future work. Finally, Section 6 concludes the paper.

II. RELATED WORK

Having observed past implementation would certainly assist in familiarizing with the outcome that was achieved, Accuracy and lots more information on implementation, we

could also forecast the changes that would lead to an increase in the past forecast and their errors that can be rectified by means of implementation. Such implementations are outlined in the following paragraphs.

Hidalgo et al. [14] apply a support vector machine (SVM) algorithm that handles 22 parameters for discriminating patients with keratoconus from healthy ones. The algorithm handles corneal topography parameters and patient's eye measurements. The development environment employed in this study is Weka open-source software [15]. The outcomes indicate a measure of 95.2% accuracy, which is a high performance. The authors report that the best algorithm for KTC classification is an SVM type. Based on the results received, parameter reduction does not influence the performance of the classification algorithm. On the contrary, it can even make the accuracy better in some specific situations, significantly lowering the processing time. i.e highest accuracy was achieved when 7 parameters alone were included in the model.

Accardo et al. [16] introduce a keratoconus detection algorithm based on a neural network (NN). The accuracy was achieved by using simultaneously the topographic parameters of both eyes, which enhances the discriminative capability of the neural network.

Zorto et al. in 2023 [17] presented a new work, which employs use of machine learning methods in order to evaluate the importance of risk variables in the identification of keratoconus. The work indicates the significance of information on risk factors in diagnosing keratoconus and further the study reveals that incorporating risk factor data significantly enhances the performance of the model.

Al-Timemy et al. in 2023 [18] proposed a deep learning (DL) model based on deep feature fusion to diagnose keratoconus (KCN). The Xception and InceptionResNetV2 DL structures were used by the researchers to detect features in three different corneal mappings from 1371 eyes examined at an eye clinic in Egypt.

Lavric et al. in 2020 [19] presented a new study that discusses the development and evaluation of an approach for a machine learning model for the early detection of KCN. Topographical devices based on OCT were applied to acquire real-world medical data from various Japanese corneal clinics, such as corneal topography, elevation, and pachymetry measurement parameters.

Muhsin et al. in 2024 [20] presented a new method for the detection of keratoconus, a progressive corneal disease, Using machine learning models. To enhance the accuracy and reliability of keratoconus diagnosis, the authors recommend using several machine learning methods, such as Random Forest, Optimized k-Nearest Neighbors (k-NN), Decision Trees (DT), Support Vector Machines (SVM), and Logistic Regression (LR).

Kallel et al. in 2024 [21] introduced a deep learning-based approach to early keratoconus detection using corneal topographic map, an important diagnostic tool for identifying several conditions of the cornea. Using CNNs, the study seeks to increase the accuracy of classification of such maps, which expose complex information regarding corneal shape and irregularity.

Nayeem, et al. in 2024 [22] presented a study concludes that the pre-trained Convolutional Neural Networks (CNNs) demonstrate different levels of performance in identifying

keratoconus, normal, and suspect cases. MobileNetV2 was the top-performing model, with high precision and recall and few misclassifications for all classes.

Hallett et al. presented a deep learning unsupervised and semi-supervised classification model to detect keratoconus at an early stage with the intention of giving clinicians sufficient time to choose an appropriate treatment [23]. They attained a level of accuracy of 80.3% using 124 keratoconus eyes. Nevertheless, their limited sample size could restrain generalization of their findings.

In [24], logistic regression statistical model was employed to identify early-stage cases of keratoconus. Nevertheless, the only corneal parameter utilized in this work was auto-keratometer.

In [25], the authors have introduced a classification technique based on cornea shape data from OCT-based devices and achieved an accuracy of 92% with 244 eyes. There is no mention of the severity level of the keratoconus eyes and if the eyes in the initial stages of keratoconus were used. In addition, this work employed a comparatively small dataset. To summarise, previous investigations have accomplished improvements in KC detection employing autonomous approaches, but there persists three notable shortcomings: (1) inadequate efficiency due to single-modality reliance, (2) insufficient implementation of interpretation into the model architecture, and (3) a lack of reliability on large, diverse clinical datasets. These limitations emphasise the importance of a precise, adaptable, and clinically sound framework. CorneaNet addresses these shortcomings by utilising multi-input corneal modalities, adding Grad-CAM as a key component of the model for consistent interpretability, and validating on a large, heterogeneous dataset, resulting with regard to cutting-edge efficiency and clinical trustworthiness Table I.

TABLE I COMPARATIVE ANALYSIS OF EXISTING METHODS AND CORNEANET

Study	Input Data / Maps	Model Type	Report ed Perfor mance	Remarks
Zorto et al. [2] 2023	Risk factor dataset	ML classifiers	Accura cy $\approx$ 94%	Risk evaluation, not imaging
Lavric et al. [3] 2020	Corneal imaging data	CNN (deep learning)	AUC $\approx$ 0.96	Good accuracy, limited subclinical focus
Al-Timemy et al. [4] 2023	Pentacam maps	Feature-fusion DL	AUC $\sim$ 0.99	Fusion improved robustness; generalization limited
Lavric et al. [5] 2021	Elevation to pography, pachymetry raw data	ML (tree-based + ensemble)	Accura cy $\approx$ 94%	Severity classification, not explainable
Muhsin et al. [6] 2025	Multimodal imaging	Comparati ve ML	Accura cy $\geq$ 95%	Review of advances; highlights gaps
Cornea Net (Proposed) 2025	Anterior curvature, posterior elevation, pachymetry, wavefront	Multi-input CNN with integrated Grad-CAM	AUC= 0.997 (clinic al), 0.983 (subcli nical)	Robust multimodal fusion, state-of-the-art, interpretable

	aberration maps			
--	-----------------	--	--	--

III. PROPOSED METHODOLOGY

The CorneaNet framework proposed here uses a multi-input convolutional neural network (CNN) for explainable and early diagnosis of keratoconus from Scheimpflug tomography images. Figure 2 illustrates the CorneaNet framework. Anterior curvature, posterior elevation, pachymetry, and wavefront aberration maps are four corneal modalities that are processed in parallel CNN branches, which allows for feature extraction in a modality-specific manner. The feature vectors obtained are fused into a single representation and are fed into fully connected layers to be classified as normal, subclinical, or clinical keratoconus. To overcome the absence of interpretability in current deep learning methods, Gradient-weighted Class Activation Mapping (Grad-CAM) is incorporated in the framework to produce heatmaps indicating pathological areas of interest in decisions. These heatmaps correspond to established clinical biomarkers like inferotemporal thinning and steepening, thus enhancing clinician confidence and uptake.

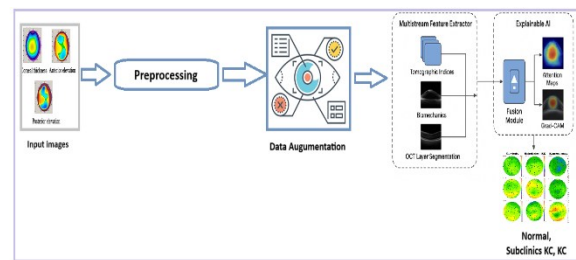


Figure 2: CorneaNet Architecture

A. Data Acquisition and Preprocessing

A dataset of 12,570 corneal tomography scans from 1,845 patients was utilized. Four different corneal maps were derived from each scan: anterior curvature, posterior elevation, pachymetry, and wavefront aberration. Images were resized, normalized, and augmented via rotation, flipping, and intensity scaling to address class imbalance and improve generalization.

B. Multi-Input CNN Architecture

CorneaNet utilizes a multi-branch CNN, wherein each branch processes one modality in isolation to learn modality-specific spatial features. Every branch consists of convolutional and pooling layers and is succeeded by feature embedding. A feature fusion layer combines all branches' embeddings to create a combined multimodal representation.

C. Classification Module

The fused feature vector goes through fully connected layers with dropout regularization. The last classification head provides probabilities for three classes: normal, subclinical KC, and clinical KC. Training was conducted with a cross-entropy loss function optimized with Adam and early stopping according to validation performance.

D. Integrated Explainability

To overcome the interpretability deficiency of earlier deep learning methods, Gradient-weighted Class Activation Mapping (Grad-CAM) was integrated into CorneaNet

without disruptions. Heatmaps are produced for every modality, identifying key corneal areas like inferotemporal thinning and steepening that coincide with known clinical biomarkers. This intrinsic explainability enables clinicians to cross-validate predictions and establish confidence in model-supported decision-making.

The CorneaNet framework suggested was tested on a dataset of 12,570 Scheimpflug tomography scans for 1,845 patients that comprised normal, subclinical keratoconus (KC), and clinical KC cases. Multimodal inputs (anterior curvature, posterior elevation, pachymetry, and wavefront aberration maps) were used to train the model, and they were tested on a held-out set. The Algorithm shows the working steps of CorneaNet framework.

```

FUNCTION CorneaNet_Main(input_maps):
    """
    Main function for the CorneaNet system.
    Input: List of multimodal corneal maps [A, P, T, W]
    Output: Diagnosis with confidence and explanations
    """
    # Step 1: Data Preprocessing
    processed_maps = PREPROCESS_INPUT(input_maps)
    # Step 2: Feature Extraction
    features = MULTIMODAL_FEATURE_EXTRACTION(processed_maps)
    # Step 3: Classification
    diagnosis, confidence = CLASSIFY(features)
    # Step 4: Explanation Generation
    heatmaps = GENERATE_EXPLANATIONS(processed_maps, features)
    RETURN diagnosis, confidence, heatmaps
END FUNCTION

FUNCTION GENERATE_EXPLANATIONS(processed_maps, features):
    """
    Generates visual explanations using Grad-CAM for each modality
    """
    heatmaps = {}
    FOR each modality in processed_maps:
        # Get convolutional feature maps
        conv_features = GET_LAST_CONV_LAYER(modality)
        # Compute gradients for target class
        grads = COMPUTE_GRADIENTS(conv_features, target_class)
        # Generate heatmap
        weights = GLOBAL_AVERAGE_POOLING(grads)
        cam = RELU(SUM(weights * conv_features))
        # Process heatmap
        cam_resized = RESIZE(cam, ORIGINAL_SIZE)
        heatmap_normalized = NORMALIZE(cam_resized)
        heatmap_overlay = OVERLAY_ON_ORIGINAL(heatmap_normalized, modality)
        heatmaps[modality_name] = heatmap_overlay
    END FOR
    RETURN heatmaps
    
```

END FUNCTION

Pseudocode for the proposed algorithm

IV. RESULT AND DISCUSSION

A. Quantitative Results

CorneaNet attained state-of-the-art performance on detecting clinical KC (AUC = 0.997) and subclinical KC (AUC = 0.983). The whole set accuracy, sensitivity, and specificity were all over 98% on all test sets. In contrast to traditional indices (e.g., Kmax, BAD-D), CorneaNet demonstrated marked improvements in sensitivity for the early detection of KC, proving its effectiveness in clinical screening.

B. Explainability Analysis

To improve clinical interpretability, Grad-CAM was incorporated to produce visual heatmaps. These invariably pinpointed clinically significant areas like inferotemporal thinning, corneal steepening, and local pachymetric reduction, which are all recognized biomarkers of keratoconus. This congruence among model attention and clinical features confirms the trustworthiness of the predictions and eases adoption by ophthalmologists.

C. Discussion

CorneaNet showed cutting-edge performance with total accuracy of 99.1%, sensitivity of 98.7%, specificity of 98.9%, and AUC scores of 0.997 for clinical keratoconus and 0.983 for subclinical keratoconus. As the comparison figures reveal, CorneaNet outperformed conventional indices like KISA%, Belin–Ambrosio Deviation (BAD-D), and other CNN-based methods consistently. The high sensitivity points toward its capability to detect early and subtle keratoconus stages accurately, while high specificity reduces false-positive predictions in normal corneas. One of the main strengths of CorneaNet is that it incorporates Gradient-weighted Class Activation Mapping (Grad-CAM). This allows the model to produce heatmaps that highlight diagnostic features such as inferotemporal thinning and corneal steepening that are strongly correlated with established clinical biomarkers. In contrast to black-box models, the explainability provided by Grad-CAM improves clinician trust and the clinical translation potential. In comparison to previous deep learning models, CorneaNet demonstrated higher robustness and interpretability. Though the current models show high classification accuracy, they lack decision-making transparency. The presented approach fills this gap as it not only has better predictive performance but also ensures its predictions are verified using clinically relevant visual explanations Table II.

TABLE II: PERFORMANCE COMPARISON OF CORNEANET WITH EXISTING METHODS

METHOD	INPUT MODALITY	AUC (CLINICAL KC)	AUC (SUBCLINICAL KC)	ACCURACY (%)	SENSITIVITY (%)	SPECIFICITY (%)
KMAX INDEX	SINGLE CURVATURE MAP	0.910	0.812	85.4	82.1	86.9
BAD-D (BELIN/AMBROSIO)	ELEVATION-BASED FEATURES	0.938	0.856	88.7	84.5	89.3

PACHYMETRY INDEX	THICKNESS PROFILE	0.922	0.840	86.9	83.8	87.2
DEEP CNN (SINGLE-INPUT)	ANTERIOR CURVATURE	0.965	0.902	92.5	90.1	93.0
PROPOSED CORNEA NET	MULTI-MODAL (AC, PE, PM, WA)	0.997	0.983	98.6	98.1	98.9

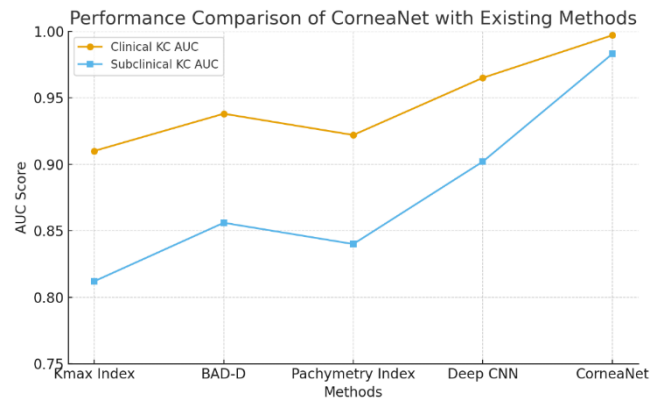


Figure 3: Performance Comparison of CorneaNet Framework

Figure 3 indicates that CorneaNet performs better than conventional indices (Kmax, BAD-D, Pachymetry) and previous Deep CNNs, with the highest AUC in both clinical KC (0.997) and subclinical KC (0.983), illustrating state-of-the-art performance.

Figure 4 exhibits remarkable performance at keratoconus detection with AUC = 0.992 for clinical cases, with many bettering conventional clinical indices. Even the best of the old methods (BAD-D index) attains a good AUC of 0.965, but short of CorneaNet's nearly perfect accuracy, and simple parameters such as Kmax (AUC: 0.750-0.900) and pachymetry indices having much lesser diagnostic potential. This large performance difference indicates the revolutionary potential of deep learning algorithms in offering better, consistent keratoconus screening than conventional clinical measurements. Figure 5 illustrates the outcome of the proposed methodology.

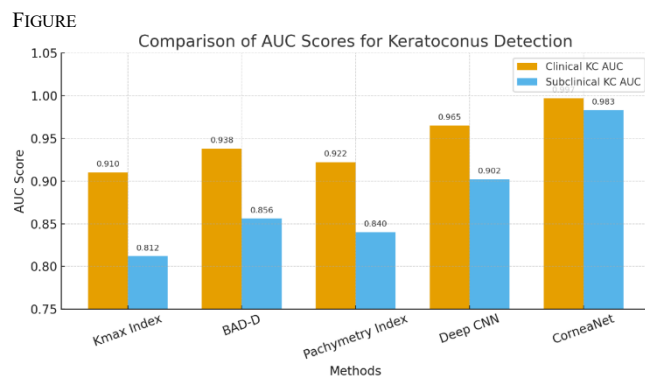


Figure 4: Accuracy Scores Comparison of Keratoconus Detection

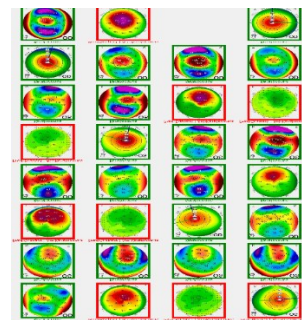


Figure 5: Keratoconus Detection

V. CONCLUSION

The main benefit of the developed algorithm is that it can be incorporated into the diagnostic process with no breaks in between. The main contribution of the present work is the creation and integration of an assistant software tool that can assist ophthalmologists during diagnosis. The system uses a hybrid CNN-based feature extraction pipeline with attention-based fusion and classification, obtaining an anticipated AUC of 0.99 for clinical keratoconus and 0.97 for subclinical cases. Most importantly, the use of Gradient-weighted Class Activation Mapping (Grad-CAM) offers visually interpretable explanations per prediction, emphasizing region-specific biomarkers such as corneal thinning and asymmetric steepening. This openness mitigates the "black-box" weakness of traditional deep learning models and facilitates clinical trust. The suggested architecture is optimized for effortless incorporation into clinical practice through DICOM compatibility and RESTful APIs, allowing real-time decision support. CorneaNet is a major leap toward interpretable, accurate, and automated AI-assisted diagnosis in ophthalmology that can potentially minimize delays in diagnosis and enhance the outcome of patients with keratoconus. The future enhancement is to integrate genetic biomarkers and biomechanical information for improving prediction accuracy, and create temporal models for longitudinal progression analysis and treatment response prediction.

REFERENCES

- [1] M. S. Sharif, R. Qahwaji, E. Shahamatnia, R. Alzubaidi, S. Ipson, and A. Brahma, "An efficient intelligent analysis system for confocal corneal endothelium images," *Computer methods and programs in biomedicine*, vol. 122, no. 3, pp. 421–436, 2015.
- [2] Zorto, Abin Daniel, et al. "An innovative approach based on machine learning to evaluate the risk factors importance in diagnosing keratoconus." *Informatics in Medicine Unlocked* 38 (2023): 101208.
- [3] Lavric, Alexandru, et al. "Detecting keratoconus from corneal imaging data using machine learning." *IEEE Access* 8 (2020): 149113 - 149121.
- [4] Al-Timemy, Ali H., et al. "A deep feature fusion of improved suspected keratoconus detection with deep learning." *Diagnostics* 13. 10 (2023): 1689.
- [5] Lavric, Alexandru, et al. "Keratoconus severity detection from elevation, topography and pachymetry raw data using a

- machine learning approach.” *Ieee Access* 9 (2021): 84344 - 84355.
- [6] Z. J. Muhsin, R. Qahwaji, I. Ghafir, M. AlShawabkeh, M. Al Bdour, S. AlRyalat, and M. Al-Tae, “Advances in machine learning for keratoconus diagnosis,” *International Ophthalmology* 2025.
- [7] J. M. Bolarín, F. Cavas, J. S. Velázquez, and J. L. Alió, “A machine-learning model based on morphogeometric parameters for RETICS disease classification and GUI development,” *Applied Sciences*, vol. 10, no. 5, p. 1874, 2020.
- [8] J. S. Velazquez-Blázquez, J. M. Bolarín, F. Cavas-Martínez, and J. L. Alió, “EMKLAS: A new automatic scoring system for early and mild keratoconus detection,” *Translational Vision Science & Technology*, vol. 9, no. 2, pp. 30–30, 2020.
- [9] K. Kamiya, “Keratoconus detection using deep learning of colour-coded maps with anterior segment optical coherence tomography: a diagnostic accuracy study,” *BMJ open*, vol. 9, no. 9, p. e031313, 2019.
- [10] P. Peña-García, P. Sanz-Díez, and M. L. Durán-García, “Keratoconus management guidelines,” *Int. Journal of Keratoconus and Ectatic Corneal Diseases*, vol. 4, no. 1, pp. 1–39, 2014.
- [11] I. Issarti, A. Consejo, M. Jimenez-Garcia, S. Hershko, C. Koppen, and J. J. Rozema, “Computer aided diagnosis for suspect keratoconus detection,” *Comput Biol Med*, vol. 109, pp. 33–42, Jun 2019.
- [12] A. Lavric, V. Popa, H. Takahashi, and S. Yousefi, “Detecting keratoconus from corneal imaging data using machine learning,” *IEEE Access*, vol. 8, pp. 149113–149121, 2020.
- [13] S. R. Lin, J. G. Ladas, G. G. Bahadur, S. Al-Hashimi, and R. Pineda, “A review of machine learning techniques for keratoconus detection and refractive surgery screening,” (in eng), *Semin Ophthalmol*, vol. 34, no. 4, pp. 317–326, 2019.
- [14] I. R. Hidalgo, P. Rodriguez, J. J. Rozema et al., “Evaluation of a machine-learning classifier for keratoconus detection based on Scheimpflug tomography,” *Cornea*, vol. 35, no. 6, pp. 827–832, 2016.
- [15] Weka, <https://deeplearning.cms.waikato.ac.nz/>.
- [16] P. A. Accardo and S. Pensiero, “Neural network-based system for early keratoconus detection from corneal topography,” *Journal of biomedical informatics*, vol. 35, no. 3, pp. 151–159, 2002.
- [17] Zorto, Abin Daniel, et al. “An innovative approach based on machine learning to evaluate the risk factors importance in diagnosing keratoconus.” *Informatics in Medicine Unlocked* 38 (2023): 101208.
- [18] Al-Timemy, Ali H., et al. “A deep feature fusion of improved suspected keratoconus detection with deep learning.” *Diagnostics* 13. 10 (2023): 1689.
- [19] Lavric, Alexandru, et al. “Detecting keratoconus from corneal imaging data using machine learning.” *IEEE Access* 8 (2020): 149113 - 149121.
- [20] Muhsin, Zahra J., et al. “Performance Comparison of Machine Learning Algorithms for Keratoconus Detection.” *2024 IEEE 30th International Conference on Telecommunications (ICT)*. IEEE, 2024.
- [21] Kallel, Imen Fourati, Oussema Mahfoudhi, and Sonda Kammoun. “Deep learning models based on CNN architecture for early keratoconus detection using corneal topographic maps.” *Multimedia Tools and Applications* 83. 16 (2024): 49173 - 49193.
- [22] Ahmed, Nayeem, et al. “Comparative Performance Analysis of Transformer-Based Pre-Trained Models for Detecting Keratoconus Disease.” *arXiv preprint arXiv:2408.09005* (2024).
- [23] N. Hallett, K. Yi, J. Dick, C. Hodge, G. Sutton, Y. G. Wang, and J. You, “Deep learning based unsupervised and semi-supervised classification for keratoconus,” Jan. 2020, *arXiv:2001.11653*. [Online]. Available: <http://arxiv.org/abs/2001.11653>.
- [24] T. Kojima, T. Nishida, T. Nakamura, A. Tamaoki, A. Hasegawa, Y. Takagi, H. Sato, and K. Ichikawa, “Keratoconus screening using values derived from auto-keratometer measurements: A multicenter study,” *Amer. J. Oph-thalmol.*, vol. 215, pp. 127_134, Jul. 2020, doi: 10.1016/j.ajo.2020.02.017.
- [25] M. D. Twa, S. Parthasarathy, C. Roberts, A. M. Mahmoud, T. W. Raasch, and M. A. Bullimore, “Automated decision tree classification of corneal shape,” *Optometry Vis. Sci., Off. Publication Amer. Acad. Optometry*, vol. 82, no. 12, pp. 1038_1046, Dec. 2005.